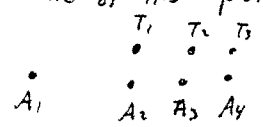


Sequences

Xeno One of his paradoxes was Achilles and the tortoise.



How can Achilles ever catch him?

Answer: Both sequences have same limit.

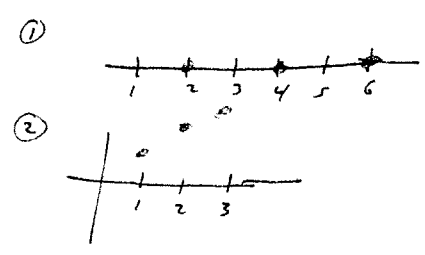
Defn A sequence is a function whose domain is $\mathbb{N} = \{1, 2, 3, \dots\}$.

Brook Defn A sequence is a list.

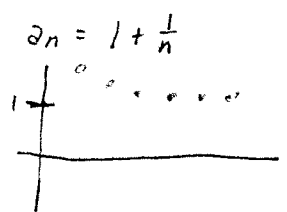
Ways to write a sequence

- ① $2, 4, 6, 8, \dots$
- ② $a_n = 2n$
- ③ $\{2n\}$ (or $\{a_n\}$)

Ways to graph a sequence



Limits A more interesting sequence is one that "goes somewhere"



converges to 1.

If $\lim_{n \rightarrow \infty} a_n$ exists, the sequence converges. Otherwise diverges.
Also write $a_n \rightarrow L$ as $n \rightarrow \infty$.

Because a sequence is really a function, all the old limit laws still hold.

Thm Suppose $a_n \rightarrow L$ and $b_n \rightarrow M$ as $n \rightarrow \infty$. Then

$$\textcircled{1} \lim_{n \rightarrow \infty} r a_n + s b_n = rL + sM$$

$$\textcircled{2} \lim_{n \rightarrow \infty} a_n b_n = LM$$

$$\textcircled{3} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{L}{M}, \text{ if } M \neq 0$$

$$\textcircled{4} \lim_{n \rightarrow \infty} \sqrt{a_n} = \sqrt{L} \quad (\text{if } \sqrt{L} \text{ exists})$$

Eg $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

Eg $\lim_{n \rightarrow \infty} \frac{2n^3 + 4n}{3n^3 + 6n^2 + 1} = \frac{2}{3}$ Why? LHR

Eg $\lim_{n \rightarrow \infty} \frac{n^4 + 7n}{3n^2 - 4} = \infty$

Tower of Power $e^n \quad n^{100} \quad n^2 \quad n \quad \sqrt{n} \quad \sqrt[3]{n} \dots \ln(n)$

Eg $\lim_{n \rightarrow \infty} \frac{n^2 \ln n}{e^n + \sqrt{n}} = 0$

Important You must have a "feel" for the "tower"

Squeeze Thm Suppose $a_n \leq b_n \leq c_n$ and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$.
Then $\lim_{n \rightarrow \infty} b_n = L$.

Eg $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$ $0 < \frac{n(n-1)\dots 1}{n \cdot n \dots n} \leq 1 \cdot 1 \dots \frac{1}{n}$, apply Squeeze Thm

Warning Can't d. LHR on $n!$

Defn a_n is increasing if $a_{n+1} \geq a_n \forall n$ (strictly if $a_{n+1} > a_n$)
" decreasing " $\leq$ " " $<$ "
 a_n is monotone if it is either increasing or decreasing

Defn a_n is bdd above by M if $a_n \leq M$
" bdd below by m " $a_n \geq m$
 a_n is bdd if it is bounded both above and below

Bounded Monotone Conv. Thm A monotone sequence converges if and only if it is bounded.

Eg $a_n = \left(\frac{1}{2}\right)^n$ Note it is dec, bdd below by 0. $\therefore$ conv. to something.
Easy to show $r^n \rightarrow 0 \quad \forall |r| < 1.$

Series

Zeno's Paradox You can't actually get to the end of the table b/c you have to do half, then half that, etc etc.
 $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ Can you add infinitely many numbers?

Brook Defn A series is a sequence added up

Defn A series $\sum_{k=1}^{\infty} a_k$ converges if the sequence of partial sums $S_n = \sum_{k=1}^n a_k$ does.

Eg $\sum_{k=1}^n k^2 = 1 + 4 + 9 + \dots$ diverges

Defn $\sum_{k=0}^{\infty} ar^k = a + ar + ar^2 + \dots$ is a geometric series

Thm $\sum_{k=0}^{\infty} ar^k$ diverges if $|r| \geq 1$ and conv. to $\frac{a}{1-r}$ if $|r| < 1$

Proof: $S_n = a + ar + \dots + ar^n$
 $-rS_n = -ar - \dots - ar^n - ar^{n+1}$
 $(1-r)S_n = a - ar^{n+1}$
 $S_n = \frac{a - ar^{n+1}}{1-r} \rightarrow \frac{a}{1-r}$

Eg $\sum_{k=0}^{\infty} \left(\frac{2}{3}\right)^k = \frac{1}{1 - 2/3} = 3$

Eg $\sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = \frac{1}{2} + \frac{1}{4} + \dots$, ~~Notice~~ Notice $\sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = \frac{1}{1 - 1/2} = 2$, so $= 1 + \frac{1}{2} + \frac{1}{4} + \dots$

$\sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k = 1$. Solves Zeno's Paradox

$$\text{Eg } \sum_{k=0}^{\infty} 4 \frac{2^{k+1}}{3^k} = \sum_{k=0}^{\infty} 8 \left(\frac{2}{3}\right)^k = 8 \frac{1}{1-\frac{2}{3}} = 24$$

$$\text{Eg } \sum_{k=2}^{\infty} (-1)^k \frac{4^{k+1}}{5^{k-1}} = \sum_{k=2}^{\infty} 20 \left(\frac{-4}{5}\right)^k$$

$$20 + 20 \left(\frac{-4}{5}\right) + \sum_{k=2}^{\infty} 20 \left(\frac{-4}{5}\right)^k = \frac{20}{1 - (-\frac{4}{5})} = 20 \left(\frac{5}{9}\right) = \frac{100}{9}$$

$$\text{So } \sum_{k=2}^{\infty} 20 \left(\frac{-4}{5}\right)^k = 20 \left(\frac{5}{9}\right) - 20 + 20 \left(\frac{4}{5}\right) = 20 \left(\frac{11}{45}\right)$$

$$\text{Eg } \sum_{k=1}^{\infty} 7 \frac{3^k}{2^{k-1}} = \sum_{k=1}^{\infty} 14 \left(\frac{3}{2}\right)^k \quad \text{diverges}$$

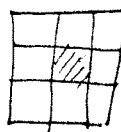
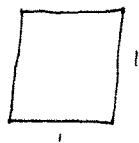
Warning We'll spend most of the chapter trying to tell if series conv or diverge, but only in a few special cases can we find the limit

Telescoping Series $\sum_{k=1}^{\infty} \frac{1}{n(n+1)}$ Recall Partial Fractions $\sum_{k=1}^{\infty} \frac{1}{k} - \frac{1}{k+1}$

$$\text{Partial sum } S_n = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1} \rightarrow 1$$

Fractals

Sierpinski Carpet



repeat

At 1st stage $\frac{1}{9}$ is colored, At 2nd stage, $\frac{1}{9}$ of $\frac{8}{9}$, so $(\frac{1}{9})(\frac{8}{9})$. Then $\frac{1}{9}$ of $\frac{1}{9}$

$$\sum_{k=1}^{\infty} \frac{1}{9} \left(\frac{8}{9}\right)^k = \frac{\frac{1}{9}}{1 - \frac{8}{9}} = 1$$

Divergence Test, Integral Test

Divergence Test If the sequence $\{a_n\}$ does not ~~go~~^{conv} to 0, then the series $\sum_{k=1}^{\infty} a_k$ must diverge.

Eg $\sum_{k=1}^{\infty} \frac{k}{2k+1}$ diverges

Warning You can never say $\sum a_k$ Converges b/c of Divergence Test

Integral Test Suppose $a_k = f(k)$ for $k=1, 2, \dots$ and f is a positive, decreasing function. Then $\sum a_k$ and $\int_1^{\infty} f(x) dx$ either both converge or both diverge.

Idea $\sum a_k$ is "like" a Riemann Sum for $\int_1^{\infty} f(x) dx$

Eg $\sum_{k=1}^{\infty} \frac{1}{k^2}$ conv. b/c $\int_1^{\infty} \frac{1}{x^2} dx = \lim_{N \rightarrow \infty} -\frac{1}{N} - (-1) = 1$

Warning: This does not mean $\sum \frac{1}{k^2}$ conv to 1 (it doesn't), but at least we know it must converge.

Harmonic Series $\sum \frac{1}{k}$ diverges b/c $\int_1^{\infty} \frac{1}{x} dx = \lim_{N \rightarrow \infty} \ln N = \infty$

p-Series $\sum \frac{1}{k^p}$ conv. iff $p > 1$

Eg $\sum \frac{1}{k \ln k}$ diverges b/c $\int_1^{\infty} \frac{dx}{x \ln x} = \lim_{N \rightarrow \infty} \ln |\ln N| = \infty$

Eg $\sum \frac{1}{k (\ln k)^{1.1}}$ conv. b/c $\int_1^{\infty} \frac{dx}{x (\ln x)^{1.1}} = \lim_{N \rightarrow \infty} \frac{-1}{N^{.1}} + .1 = .1$

Comparison Tests

Direct Comparison Test (Baby Comparison Test) Suppose $0 \leq a_n \leq b_n \forall n$.

- ① $\sum b_n$ conv $\Rightarrow \sum a_n$ conv
- ② $\sum a_n$ diverges $\Rightarrow \sum b_n$ diverges

Idea  Is the big one small enough to converge?
Is the small one too big?

Aside Terms must be positive. $\frac{1}{n} < \frac{1}{n^2}$, but $\sum \frac{1}{n}$ diverges

Eg 1 $\sum \frac{\ln n}{n}$ diverges b/c $\frac{\ln n}{n} > \frac{1}{n}$ and $\sum \frac{1}{n}$ div

Eg 2 $\sum_{k=1}^{\infty} \frac{3}{4^k + 5}$ conv. b/c $\frac{3}{4^k + 5} < 3 \left(\frac{1}{4}\right)^k$

Eg 3 $\sum_{k=2}^{\infty} \frac{1}{4\sqrt{k} - 3}$ div. b/c $\frac{1}{4\sqrt{k} - 3} > \frac{1}{4\sqrt{k}}$

Idea We are really thinking $\frac{3}{4^k + 5} \approx \frac{3}{4^k}$ and $\frac{1}{\sqrt{k} - 3} \approx \frac{1}{\sqrt{k}}$

But the Baby Test doesn't really help much with $\frac{1}{4\sqrt{k} + 3} < \frac{1}{4\sqrt{k}}$!!

Limit Comparison Test Suppose $0 < a_k, b_k$ and $\lim_{k \rightarrow \infty} \frac{a_k}{b_k}$ is not

0 or ∞ , then $\sum a_k$ and $\sum b_k$ either both conv or both diverge.

Eg $\sum \frac{1}{\sqrt{k}+3}$ looks like $\sum \frac{1}{\sqrt{k}}$

$$\lim_{k \rightarrow \infty} \frac{\frac{1}{\sqrt{k}+3}}{\frac{1}{\sqrt{k}}} = \lim_{k \rightarrow \infty} \frac{\sqrt{k}}{\sqrt{k}+3} = 1 \quad \text{both diverge}$$

Eg $\sum \frac{3}{4^k - 5}$ looks like $\sum 3\left(\frac{1}{4}\right)^k$

$$\lim_{k \rightarrow \infty} \frac{\frac{3}{4^k - 5}}{3\left(\frac{1}{4}\right)^k} = \lim_{k \rightarrow \infty} \frac{4^k}{4^k - 5} = 1 \quad \text{both converge}$$

Eg $\sum \frac{5}{2k^2 - 4k + 3}$ looks like $\sum \frac{1}{k^2}$

$$\lim_{k \rightarrow \infty} \frac{5k^2}{2k^2 - 4k + 3} = \frac{5}{2} \quad \text{both converge}$$

Eg $\sum \frac{2k^3 + 3k}{3k^4 - 5}$ looks like $\sum \frac{1}{k}$

$$\lim_{k \rightarrow \infty} \frac{2k^4 + 3k^2}{3k^4 - 5} = \frac{2}{3} \quad \text{both diverge}$$

Eg $\sum \frac{3\sqrt{k}+1}{k^2+3k}$ looks like $\sum \frac{1}{k^{1.5}}$

$$\lim_{k \rightarrow \infty} \frac{3k^{1.5} + k^{1.5}}{k^2 + 3k} = 3 \quad \text{both Converge}$$

Zero-Infinity Test (Comparison Common Sense) Suppose $0 < a_k, b_k$.

$$\textcircled{1} \lim_{k \rightarrow \infty} \frac{a_k}{b_k} = 0 \quad \text{and} \quad \sum b_k \text{ conv} \Rightarrow \sum a_k \text{ conv}$$

$$\textcircled{2} \lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \infty \quad \text{and} \quad \sum b_k \text{ div} \Rightarrow \sum a_k \text{ div.}$$

Eg $\sum \frac{\ln k}{\sqrt[3]{k}}$ $\textcircled{1}$ Compare with $\sum \frac{1}{\sqrt{k}}$

$$\lim_{k \rightarrow \infty} \frac{\frac{\ln k}{\sqrt[3]{k}}}{\frac{1}{\sqrt{k}}} = \lim_{k \rightarrow \infty} \frac{\ln k}{\sqrt[3]{k}} \cdot \sqrt{k} = \lim_{k \rightarrow \infty} k^{1/6} \ln k = \infty$$

This means $\frac{\ln k}{\sqrt[3]{k}}$ is bigger than $\frac{1}{\sqrt{k}}$, so it must diverge

$\textcircled{2}$ Direct comparison also works
Eg $\sum \frac{4k^3}{3^k}$

log-q Series $\sum \frac{\ln k}{k^q}$ conv iff $q > 1$ (Just like p-series)

Ratio Test

Ratio Test Suppose $a_k > 0$ and $\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = L$. Then

- ① $L < 1 \Rightarrow \sum a_k$ converges
- ② $L > 1 \Rightarrow \sum a_k$ diverges
- ③ $L = 1 \Rightarrow$ nothing

Eg $\sum_{k=1}^{\infty} \left(\frac{1}{3}\right)^k \quad \lim_{k \rightarrow \infty} \frac{\left(\frac{1}{3}\right)^{k+1}}{\left(\frac{1}{3}\right)^k} = \frac{1}{3} \quad \text{conv.}$

Idea The ratio test checks to see if terms are dropping off by some factor. This is great for a^k or $k!$ terms

Eg $\sum_{k=1}^{\infty} \frac{1}{k+4} \quad \lim_{k \rightarrow \infty} \frac{k+4}{k+5} = 1 \quad \text{no help, but diverges}$

Eg $\sum_{k=1}^{\infty} \frac{1}{k^2+4} \quad \lim_{k \rightarrow \infty} \frac{k^2+4}{(k+1)^2+4} = 1 \quad \text{no help, but converges}$

Eg $\sum \frac{3^k}{(k+2)!} \quad \lim_{k \rightarrow \infty} \frac{3^{k+1}}{(k+3)(k+2)\dots 1} \cdot \frac{(k+2)\dots 1}{3^k} = \lim_{k \rightarrow \infty} \frac{3}{k+3} = 0 \quad \text{conv.}$

⑧ Eg $\sum \frac{k^k}{k!} \quad \lim_{k \rightarrow \infty} \frac{(k+1)^{k+1}}{(k+1)!} \cdot \frac{k!}{k^k} = \lim_{k \rightarrow \infty} \frac{(k+1)^k}{k^k} = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k = e \quad \underline{\text{div.}}$

Aside Recall LHR for $\lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k = e$. Take ln first

$$\lim_{k \rightarrow \infty} \frac{\ln\left(1 + \frac{1}{k}\right)}{\frac{1}{k}} = \frac{1}{1 + \frac{1}{k}} \cdot \frac{(-k^{-2})}{(-k^{-2})} = 1$$

Eg Find all $x > 0 \Rightarrow \sum k^3 x^k$ converges.

Try Ratio Test. $\lim_{k \rightarrow \infty} \frac{(k+1)^3 x^{k+1}}{k^3 x^k} = x.$

So we converge if $0 \leq x < 1.$

Check $x=1$ separately. $\sum k^3$ diverges, so answer is $[0, 1).$

Eg Better example for future purposes is $\sum k^3 |x-5|^k.$

$$\lim_{k \rightarrow \infty} \frac{(k+1)^3 |x-5|^{k+1}}{k^3 |x-5|^k} = |x-5|$$

$$\begin{array}{lll} \text{Solve } |x-5| < 1. & \begin{array}{l} x-5 < 1 \\ x < 6 \end{array} & \boxed{\text{or}} \quad \begin{array}{l} x-5 > -1 \\ x > 4 \end{array} \end{array}$$

Check $x=4, 6$. Diverges. So $(4, 6).$

Root Test

Root Test Suppose $a_k > 0$ and $\lim_{k \rightarrow \infty} \sqrt[k]{a_k} = L$

- ① $L < 1 \Rightarrow \sum a_k$ converges
- ② $L > 1 \Rightarrow \sum a_k$ diverges
- ③ $L = 1 \Rightarrow$ nothing

Aside The Root Test is a little harder to apply, but more powerful

Eg $\sum 3^{-k} k^k$ $\lim_{k \rightarrow \infty} \sqrt[k]{3^{-k} k^k} = \lim_{k \rightarrow \infty} \frac{k}{3} = \infty$ diverges

Eg $\sum \frac{k^2}{2^k}$ $\lim_{k \rightarrow \infty} \sqrt[k]{\frac{k^2}{2^k}} = \lim_{k \rightarrow \infty} \frac{(\sqrt[k]{k})^k}{2} = \frac{1}{2}$ converges

Aside $\sqrt[k]{k} \rightarrow 1$ by LHR

Alternating Series

Defn In an alternating series terms alternate signs; eg, $\sum (-1)^k a_k$
or $\sum (-1)^{k+1} a_k$ with $a_k > 0$.

Alternating Series Test If $\sum (-1)^k a_k$ is alternating and
① $\{a_k\}$ is a decreasing sequence
② $\lim_{k \rightarrow \infty} a_k = 0$,
then $\sum (-1)^k a_k$ converges.

Eg $\sum (-1)^k \frac{1}{k}$ converges (to $\ln 2$)

Eg $\sum (-1)^k \frac{3k}{4k-1}$ diverges b/c of Divergence Test

Eg $\sum (-1)^{k+1} \frac{k^2}{k^3+4}$ converges by Alt. Series Test

But it "just got lucky." Without the $(-1)^{k+1}$ it would diverge.

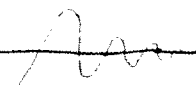
Defn $\sum a_k$ converges absolutely if $\sum |a_k|$ converges
If $\sum a_k$ converges, but not absolutely, then it converges conditionally

Eg $\sum (-1)^k \frac{1}{k^2}$ conv. abs but $\sum (-1)^k \frac{1}{k}$ conv. cond.

Eg $\sum \frac{\cos k}{k^2}$ conv. abs. because $\sum \left| \frac{\cos k}{k^2} \right|$ conv. by Baby Comp. Test w/ $\sum \frac{1}{k^2}$

Better Ratio Test Recall the Ratio Test required $a_k > 0$. But we could just check $\sum |a_k|$ to check absolute convergence.

Error Estimator Notice if $\sum (-1)^k a_k$ satisfies the Alt. Series Test, then it is "1 step forward, $\frac{1}{2}$ step back" kind of sum.

Sum  One term ^{above the limit} pushes sum up, the next pushes it below the limit.

So the difference between the partial sum and the true sum is less than the next term!

Rearranging Terms Its probably not too surprising that you can rearrange the terms in an abs. conv. series and still get the same sum, but

A conditionally convergent sum can be re-arranged to sum to anything you want.

Proof Idea The positive terms and negative terms both diverge separately. So, add up positive terms until you get past L . Subtract negative pieces until you get below L . Repeat. Since terms $\rightarrow 0$, the "overshooting" $\rightarrow 0$.

Power Series

Defn $\sum_{k=0}^{\infty} a_k (x-a)^k$ is a power series centered at a

Eg $\sum_{k=1}^{\infty} x^k$. If you plug in a number for x , you get a geometric series

It converges if $|x| < 1$.

IMPORTANT A power series is a function.

Its value and whether or not it converges depends on which x you plug in. Think of them as "long polynomials"

Thm For a power series $\sum a_k (x-a)^k$ exactly one of these possibilities occurs

① It converges only when $x=a$

② It converges for all numbers x

③ $\exists R > 0 \ni$ It converges absolutely if $|x-a| < R$
It diverges if $|x-a| > R$

R is the Radius of Convergence

Eg $\sum \frac{2^k}{k} x^k$ $\lim_{k \rightarrow \infty} \left| \frac{a_{k+1} x^{k+1}}{a_k x^k} \right| = \lim_{k \rightarrow \infty} \frac{k}{k+1} |2x| = |2x|$

By Ratio Test, the series conv. abs. when $|2x| < 1$, or $|x| < \frac{1}{2}$.

Still must check endpoints b/c Ratio Test tells us nothing there.

$x = \frac{1}{2}$ $\sum \frac{1}{k}$ diverges

$x = -\frac{1}{2}$ $\sum (-1)^k \frac{1}{k}$ conv. conditionally.

Radius of Convergence $R = \frac{1}{2}$

Interval of Convergence $[-\frac{1}{2}, \frac{1}{2})$

Eg $\sum K^K (X-2)^K$ Easiest way is Root Test

$$\lim_{K \rightarrow \infty} \sqrt[K]{|K^K (X-2)^K|} = \lim_{K \rightarrow \infty} K |X-2| = \begin{cases} 0, & X=2 \\ \infty, & X \neq 2 \end{cases}$$

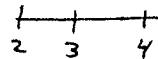
~~the rest is what X is~~

So Radius of convergence = 0 Converges only when $X=2$

Eg $\sum \frac{(X-3)^K}{K}$

$$\lim_{K \rightarrow \infty} \left| \frac{(X-3)^{K+1}}{K+1} \cdot \frac{K}{(X-3)^K} \right| = \lim_{K \rightarrow \infty} \frac{K}{K+1} |X-3| = |X-3|$$

Solve $|X-3| < 1 \Rightarrow 2 < X < 4$



Radius of Conv = 1

Check Endpoints! $X=4 \Rightarrow \sum \frac{1}{K}$ div

$X=2 \Rightarrow \sum (-1)^K \frac{1}{K}$ conv.

Interval of Convergence $[2, 4)$

$$\text{Eg } \sum \frac{x^k}{k!} \quad \lim_{k \rightarrow \infty} \left| \frac{\frac{x^{k+1}}{(k+1)!}}{\frac{x^k}{k!}} \right| = \lim_{k \rightarrow \infty} \frac{|x|}{k+1} = 0$$

This is < 1 no matter what x is. Radius of Conv $= \infty$ Int of Conv $= (-\infty, \infty)$

Calculus of Power Series If a power series is a function, can we integrate or differentiate it? How much like a polynomial is it?

$$\text{Thm } \frac{d}{dx} \sum_{k=0}^{\infty} a_k (x-a)^k = \sum_{k=1}^{\infty} k a_k (x-a)^{k-1}$$

$$\int \sum_{k=0}^{\infty} a_k (x-a)^k dx = \sum_{k=0}^{\infty} \frac{a_k}{k+1} (x-a)^{k+1}$$

This is valid on the interval of convergence.

$$\text{Eg } \frac{d}{dx} \sum_{k=0}^{\infty} \frac{x^k}{k!} = \sum_{k=1}^{\infty} \frac{k}{k!} x^{k-1} = \sum_{k=0}^{\infty} \frac{k+1}{(k+1)!} x^k = \sum_{k=0}^{\infty} \frac{1}{k!} x^k$$

$$\frac{d}{dx} \left(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots \right) = \left(1 + x + \frac{1}{2}x^2 + \dots \right)$$

So what is $\sum_{k=0}^{\infty} \frac{1}{k!} x^k$? It must be e^x !

Taylor Series

We saw that we could get e^x as a power series $\sum_{k=0}^{\infty} \frac{x^k}{k!}$.
Can we do that with other functions?

Thm If $f(x)$ can be written as a power series, it must be

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \dots$$

This is the Taylor Series for f . When $a=0$, it is called the Maclaurin series for f .

Historical Note Newton used Taylor Series all over the place when ^{Brook} Taylor was a preschooler. ~~Later~~ James Gregory did almost the same thing before both of them.

It seems odd to give Maclaurin credit for just putting $a=0$ in Taylor's series, but Maclaurin discovered Cramer's Rule first and Cramer got credit for it, so maybe it balances out.

Check Let $f(x) = \cancel{f(a)} + a_0 + a_1(x-a) + a_2(x-a)^2 + a_3(x-a)^3 + \dots$

$$f(a) = a_0 + 0 = a_0$$

$$f'(a) = f'(a) + 0 = f'(a)$$

$$f''(a) = \frac{f''(a)}{2} 2 + 0 = f''(a)$$

etc

Eg $f(x) = e^x$

$f^k(0) = e^0 = 1$

so $\sum_{k=0}^{\infty} \frac{1}{k!} x^k$

Eg $f(x) = \cos x$ Maclaurin series

$f(0) = \cos(0) = 1$

$f'(0) = -\sin(0) = 0$

$f''(0) = -\cos(0) = -1$

$f'''(0) = \sin(0) = 0$

repeats

$\cos x = \left(1 - \frac{1}{2} x^2 + \frac{1}{4!} x^4 - \frac{1}{6!} x^6 + \frac{1}{8!} x^8 - \dots \right)$

$= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k}$

Eg $f(x) = \sin x = x - \frac{1}{3!} x^3 + \frac{1}{5!} x^5 - \frac{1}{7!} x^7 + \dots$

$f(0) = \sin(0) = 0$

$f'(0) = \cos(0) = 1$

$f''(0) = -\sin(0) = 0$

$f'''(0) = -\cos(0) = -1$

etc

$\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$

Mystery Solved Why does LHR work? Why $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$?

$\lim_{x \rightarrow 0} \frac{x - \frac{1}{6} x^3 + \dots}{x} = \lim_{x \rightarrow 0} 1 - \frac{1}{6} x^2 + \dots = 1$

Euler's Formula $e^{ix} = \sum_{k=0}^{\infty} \frac{1}{k!} (ix)^k = 1 + ix - \frac{1}{2!} x^2 - i \frac{1}{3!} x^3 + \frac{1}{4!} x^4 + \dots$

$$\left(1 - \frac{1}{2!} x^2 + \frac{1}{4!} x^4 - \frac{1}{6!} x^6 + \dots \right) + i \left(x - \frac{1}{3!} x^3 + \frac{1}{5!} x^5 - \dots \right)$$

$\cos x$

$+ i \sin x$

Corollary $e^{i\pi} = -1$

$$\text{Eg } f(x) = x^3 \cos(4x) \quad \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} 4^{2k} x^{2k+3}$$

$$\text{Eg } \int e^{x^2} dx = \int \sum_{k=0}^{\infty} \frac{1}{k!} x^{2k} dx = \sum_{k=0}^{\infty} \frac{1}{(2k+1)k!} x^{2k+1}$$

$$\text{Eg } \frac{3}{x} \sin(5x^4) = \sum_{k=0}^{\infty} \frac{3(-1)^k}{(2k+1)!} 5^{2k+1} x^{8k+3}$$

$$\text{Eg } \frac{1}{1+x^2} = \frac{1}{1-(-x^2)} \quad \text{Geometric Series} \quad \sum_{k=0}^{\infty} (-1)^k x^{2k}$$

$$\text{Eg } \tan^{-1} x \text{ is the integral of the above. } \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1}$$

$$\text{Notice then } \tan^{-1}(1) = \frac{\pi}{4}, \text{ so}$$

$$\pi = 4 \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}$$

Memorize Series for e^x , $\sin x$, $\cos x$, $\frac{1}{1-x}$ and above tricks

$$\text{Eg } \sum_{k=0}^{\infty} \frac{2^k}{k!} \text{ conv. by ratio test, etc, but limit} = e^2$$